A Project Report

On

"Face Emotion Detection & Music Recommendation System"

by

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under the mentorship of

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APRIL 2026

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CERTIFICATE

This is to certify that the Project Report submitted by **JIVANI SUJAL PRAVINBHAI (**25IC03CS008**)** to the **P P SAVANI UNIVERSITY** for the partial fulfilment of the subject credit requirements is a bonafied work carried out by the student.

This is to further certify that I have been supervising the Major/Minor Project of **JIVANI SUJAL PRAVINBHAI (**25IC03CS008**).**

The contents of this report, in full or in parts, have not been submitted to any other Institute or University for award of any degree, diploma or titles.

Sign of Faculty Mentor : ____________________________

Name of Faculty Mentor : MR. SAURAV SINGH

Date: ______________________

CERTIFICATE

This is to certify that the Project Report submitted by **RAKHOLIYA KRIPA RAJUBHAI (**25IC03CS035**)** to the **P P SAVANI UNIVERSITY** for the partial fulfilment of the subject credit requirements is a bonafied work carried out by the student.

This is to further certify that I have been supervising the Major/Minor Project of **RAKHOLIYA KRIPA RAJUBHAI (**25IC03CS035**).**

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ACKNOWLEDGEMENT

This report would not have been possible without my teachers who were always there when I needed them the most. I take this chance to acknowledge them and extend my sincere gratitude for helping me make this Report a possible.

I wish to thank my faculty mentor **Mr. Saurav Singh,** Assistant Professor, School of Engineering. It has been an honor to learn under their mentorship.

As my mentor, he has constantly motivated me to remain focused on achieving my goal. Their observations and guidance helped me to establish the overall direction of the report and to move forward with learning in depth. Their vital support at each juncture, which culminated in successful completion of my project work. I express my sincere gratitude to them for constant support during the project work.

I am also thankful to faculty members of the department for constant support and guidance.

I am thankful to Dean, School of Engineering for his initiative of imparting Project during tenure of your study making us learn new things and to help us in expanding our horizons.

Name of Student : **JIVANI SUJAL PRAVINBHAI**
Enrollment No : **25IC03CS008**

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Enrollment No : **25IC03CS035**

PREFACE

In recent years, the field of Machine Learning has gained significant importance in developing intelligent systems that can understand and respond to human behavior. One of the most interesting applications of this technology is the ability to recognize human emotions through facial expressions.

This project, titled “Face Emotion Detection and Music Recommendation System”, is an effort to combine the power of Computer Vision and Deep Learning to create a smart and interactive system. The main objective of this project is to detect the emotional state of a user using facial expressions captured through a webcam and to recommend suitable music based on the detected emotion.

The development of this project has provided valuable insights into various Machine Learning concepts such as image processing, Convolutional Neural Networks (CNN), data preprocessing, and model evaluation. It has also helped in understanding how artificial intelligence can be applied in real-life scenarios to enhance user experience.

This project not only focuses on technical implementation but also highlights the importance of personalization in today’s digital world. By integrating emotion detection with music recommendation, the system aims to provide users with a more engaging and satisfying experience.

I would like to express my gratitude to all those who supported and guided me throughout the development of this project. Their encouragement and valuable suggestions played a crucial role in the successful completion of this work.

This project has been a great learning experience, and I hope it contributes to further advancements in the field of Machine Learning and intelligent systems.

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INTRODUCTION

In the modern digital world, music plays an essential role in human life and is closely connected to emotions. People often prefer listening to songs that match their current mood, as it helps in relaxation, motivation, and emotional expression. However, selecting appropriate music manually can be time-consuming and may not always reflect the user's exact emotional state.

Existing System

In the existing system, music recommendation is generally based on user preferences such as search history, playlists, or manually selected genres. Most music applications provide recommendations using basic algorithms that depend on previously played songs or user ratings.

Some platforms also use recommendation techniques like popularity-based suggestions or collaborative filtering, where users receive recommendations based on similar users' interests. However, these systems do not consider the real-time emotional state of the user.

In addition, there are some basic emotion detection systems available, but they are often limited to research environments and are not fully integrated with real-time applications like music recommendation systems.

Limitations of Existing System

The existing systems have several limitations that reduce their effectiveness in providing a personalized experience:

- **Lack of Real-Time Emotion Detection:** Current systems do not detect the user's live emotional state, which is crucial for accurate music recommendation.

- **Manual Effort Required:** Users need to manually search or select songs, which can be inconvenient and time-consuming.

- **Limited Personalization:** Recommendations are based on past behavior rather than current mood, leading to less relevant suggestions.

- **No Integration of Emotion with Recommendation:**Emotion detection and music recommendation are usually treated as separate systems and are not combined effectively.

- **Reduced User Satisfaction:** Since the system does not understand the user's emotions, it may fail to provide the desired music experience.

To overcome these limitations, the proposed system integrates real-time face emotion detection with an intelligent music recommendation mechanism, providing a more accurate, automated, and personalized solution.

PROPOSED SYSTEM

The proposed system titled **“Face Emotion Detection and Music Recommendation System”** is an advanced Machine Learning and Computer Vision-based application designed to automatically detect human emotions through facial expressions and recommend suitable music in real time. The system is developed to improve user experience by providing personalized music suggestions based on the user’s current emotional state.

In the existing systems, music recommendation is generally based on user search history, playlists, or manually selected preferences. However, these systems fail to understand the real-time emotional condition of the user. The proposed system overcomes this limitation by introducing an intelligent emotion recognition module that works in real time using facial expressions.

This system combines multiple technologies such as Machine Learning, Deep Learning, and Computer Vision to achieve accurate emotion detection and personalized music recommendation. The system captures live video input from a webcam, processes facial images, and uses a trained Convolutional Neural Network (CNN) model to classify emotions.

The major emotions considered in this system are:

- Happy
- Sad
- Angry
- Neutral
- Surprise

Based on the detected emotion, the system maps the result to a suitable category of music and provides recommendations accordingly.

2.1 PROJECT PROFILE

PROJECT NAME	FACE EMOTION DETECTION & MUSIC RECOMMENDATION SYSTEM
TECHNOLOGY	Python, OpenCV, TensorFlow/Keras, Flask
FRONTEND	HTML, CSS, JavaScript
BACKEND	Python (Flask Framework)
INTERNAL GUIDE	Mr. Saurav Singh, Assistant Professor
SUBMITTED TO	School of Engineering [P P SAVANI UNIVERCITY]
DEVELOPED BY	Sujal Jivani & Kripa Rakholiya

2.2 INTRODUCTON OF THE PROPOSED SYSTEM

The proposed system is designed to create a bridge between human emotions and digital entertainment. Music is an important part of human life and is often used to regulate mood and emotions. However, selecting the right music manually can sometimes be difficult and time-consuming.

To solve this issue, the system introduces an automated approach where emotions are detected using facial expressions. Facial expressions are one of the most natural and accurate indicators of human emotions. By analyzing these expressions, the system can understand the user's mood in real time.

The system uses Convolutional Neural Networks (CNN), which are highly effective in image classification tasks. CNN automatically extracts important features such as edges, textures, and facial patterns without manual intervention.

The workflow of the system is as follows:

1. Capture image using webcam
2. Detect face from the image
3. Preprocess the image
4. Feed image into CNN model
5. Predict emotion
6. Recommend music based on emotion

This approach ensures that the system is fully automated and does not require any manual input from the user.

The proposed system is not only useful for entertainment purposes but also has potential applications in mental health monitoring, smart assistants, and human-computer interaction systems.

2.3 BASIC FUNCTIONALITY OF THE SYSTEM

The system performs several important functions that work together to achieve the final output.

- **Real-Time Image Capture**
  The system continuously captures images from the webcam in real time. Each frame is analyzed separately to detect faces and emotions.

- **Face Detection**
  The system identifies human faces in the captured frame using computer vision techniques. Only the detected face region is considered for further processing.

- **Image Preprocessing**
  Before feeding the image into the model, preprocessing is performed. This includes:
  - Conversion to grayscale
  - Resizing to standard dimensions
  - Normalization of pixel values
  - Noise reduction (if required)

- **Emotion Classification**
  The preprocessed image is passed to the CNN model, which classifies the emotion into predefined categories. The model outputs probabilities for each emotion class, and the highest probability is selected as the final result.

- **motion Display**
  The detected emotion is displayed on the screen along with the webcam feed for real-time visualization.

- **Music Recommendation**
  Based on the detected emotion, the system recommends music from a predefined mapping. For example:
  - Happy → Energetic and party songs
  - Sad → Soft and emotional songs
  - Angry → Calm and relaxing music
  - Neutral → Random or mixed playlist
  - Surprise → Trending songs

- **User Interaction**
  The system provides a simple interface where users can view their detected emotion and listen to recommended songs, making it interactive and user-friendly.

2.4 SCOPE OF THE PROPOSED SYSTEM

The scope of this system is broad and can be extended in multiple directions.

Current Scope

The current system includes:

- Real-time emotion detection using webcam
- Classification of basic emotions
- Music recommendation based on emotion
- Single user interaction system
- Desktop-based implementation

Application Scope

The system can be applied in various fields such as:

- Entertainment industry (music apps)
- Mental health monitoring systems
- Smart assistant applications
- Human-computer interaction systems

Future Scope

The system can be further improved by adding advanced features such as:

- Integration with Spotify or YouTube Music APIs
- Mobile application development
- Multi-user emotion detection
- Voice emotion recognition
- Advanced deep learning models like ResNet or LSTM
- Personalized recommendation using AI-based learning algorithms

2.5 OBJECTIVE OF THE PROPOSED SYSTEM

The main objectives of the system are:

- To develop a real-time facial emotion detection system
- To implement a Convolutional Neural Network (CNN) for accurate classification
- To integrate emotion detection with music recommendation
- To automate the music selection process based on user mood
- To improve user experience through personalization
- To reduce manual effort in searching for music
- To demonstrate the real-world application of Machine Learning and Artificial Intelligence
- To build an efficient, scalable, and user-friendly system

The ultimate goal is to create an intelligent system that understands human emotions and responds appropriately with personalized music recommendations.

INTRODUCTIOIN OF DEVELOPMENT ENVIRONMENT

The development environment of this project consists of tools, technologies, and frameworks used to design, develop, and implement the Face Emotion Detection and Music Recommendation System. The system is mainly built using Python and integrates Machine Learning, Deep Learning, and Computer Vision concepts.

3.1 WHAT IS MACHINE LEARNING?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed. Instead of writing fixed rules, ML systems use algorithms to identify patterns in data and make predictions or decisions.

In this project, Machine Learning is used to detect human emotions from facial expressions. The system is trained on a dataset of facial images, where each image is labeled with a corresponding emotion such as happy, sad, angry, neutral, or surprise.

A Convolutional Neural Network (CNN), which is a type of deep learning algorithm, is used to extract features from images and classify emotions accurately. The model learns patterns such as facial structure, eye movement, and mouth position to predict the correct emotion.

Machine Learning plays a crucial role in making the system intelligent, adaptive, and capable of real-time decision-making.

3.2 DATABASE

A database is an organized collection of data that is stored and managed in a structured way so that it can be easily accessed, updated, and used by applications.

In this project, the database concept is used in two ways:

3.3 Dataset as a Database

The primary data used in this project is the **FER2013 dataset**, which acts as a training database for the Machine Learning model. This dataset contains thousands of facial images labeled with emotions.

The dataset is used to:

- Train the CNN model
- Validate model performance
- Test emotion classification accuracy

Each image in the dataset represents a specific emotion category, making it a structured collection of labeled data similar to a database.

3.4 Music Recommendation Data Storage

For the music recommendation system, a simple structured data storage is used to map emotions to music categories. This can be implemented using:

- Python dictionaries
- CSV files
- Local database systems (optional like SQLite or MySQL)

Example:

- Happy → Energetic songs
- Sad → Soft songs
- Angry → Calm music
- Neutral → Mixed playlist
- Surprise → Trending songs

This mapping acts as a lightweight database that stores relationships between emotions and music types.

PROBLEM STATEMENT

In today's digital era, music has become an integral part of human life and plays a significant role in influencing emotions, mood, and mental well-being. People often rely on music for relaxation, motivation, stress relief, and emotional expression. However, selecting the right type of music according to one's current emotional state remains a challenge. Most users manually search for songs or playlists, which may not always match their exact mood at a given moment.

Existing music recommendation systems, such as those used in popular streaming platforms, are primarily based on user behavior, listening history, preferences, ratings, and collaborative filtering techniques. While these systems provide personalized suggestions, they lack the ability to understand the user's real-time emotional condition. As a result, the recommendations generated may not always align with the user's present mood, leading to reduced satisfaction and engagement.

Another major limitation of current systems is the absence of real-time emotion detection. Human emotions are dynamic and can change frequently depending on various factors such as environment, stress, and personal experiences. Traditional systems fail to capture these changes instantly, as they depend only on past data rather than current emotional expressions.

Although research has been conducted in the field of emotion detection using facial expressions, most of these systems are limited to academic or experimental environments and are not effectively integrated into real-world applications like music recommendation systems. There is a lack of systems that combine emotion recognition with intelligent decision-making to provide meaningful outputs.

Furthermore, manual selection of music requires time and effort, which can be inconvenient for users, especially when they are unable to clearly identify or express their own emotions. This highlights the need for an automated system that can detect emotions without requiring user input and provide suitable recommendations instantly.

Therefore, the main problem addressed in this project is the development of an intelligent system that can automatically detect human emotions in real time using facial expressions and provide appropriate music recommendations based on the detected emotion. The system aims to bridge the gap between human emotional understanding and digital music services by integrating Machine Learning, Deep Learning, and Computer Vision techniques.

OBJECTIVES

The main objective of the project titled ***"Face Emotion Detection and Music Recommendation System"*** is to develop an intelligent and automated system that can understand human emotions in real time and provide personalized music recommendations accordingly. The project aims to combine the concepts of Machine Learning, Deep Learning, and Computer Vision to enhance user experience and reduce manual effort in music selection.

The specific objectives of this project are as follows:

- **To detect human emotions using facial expressions**

The system aims to capture real-time facial images through a webcam and analyze them to identify human emotions. Facial expressions are one of the most natural indicators of emotions, and detecting them accurately is the first step toward building an emotion-aware system.

- **To develop a CNN-based emotion classification model**

Another important objective is to design and implement a Convolutional Neural Network (CNN) model that can effectively classify emotions such as happy, sad, angry, neutral, and surprise. The model should be trained on a suitable dataset to achieve high accuracy and reliable performance.

- **To recommend music based on detected emotions**

The system is designed to suggest appropriate music according to the detected emotional state of the user. Each emotion is mapped to a specific category of songs, ensuring that the recommendations match the user's current mood.

- **To automate the music selection process**

The project aims to eliminate the need for manual searching and selection of songs. By automating the entire process—from emotion detection to music recommendation—the system provides a seamless and efficient user experience.

- **To improve user experience through personalization**

One of the key objectives is to enhance user satisfaction by delivering personalized content. By understanding the user’s emotions in real time, the system provides more relevant and meaningful music recommendations, making the interaction more engaging and enjoyable.

SYSTEM PLANNING

System planning is an important phase in the development of the *Face Emotion Detection and Music Recommendation System*. It includes analyzing feasibility, managing risks, creating a project plan, and selecting an appropriate process model to ensure successful system development.

6.1 FEASIBILTY STUDY

The feasibility study is conducted to evaluate whether the proposed system is practical and achievable. It includes technical, economic, and operational feasibility.

- **Technical Feasibility:**
  The system is technically feasible as it uses widely available technologies such as Python, OpenCV, and TensorFlow. The required hardware, such as a webcam and a standard computer system, is easily accessible. The use of pre-trained models and datasets like FER2013 further supports implementation.

- **Economic Feasibility:**
  The project is cost-effective as it uses open-source tools and software. No expensive hardware or paid software is required, making it suitable for academic purposes.

- **Operational Feasibility:**
  The system is user-friendly and easy to operate. It requires minimal user interaction and provides real-time results, making it practical for everyday use.

6.2 RISK MANAGEMENT

Risk management involves identifying potential risks and taking measures to reduce their impact on the project.

Some possible risks in this project include:

- **Accuracy Issues:** Emotion detection may not always be accurate due to lighting conditions or facial variations.
- **Hardware Dependency:** Poor camera quality can affect system performance.
- **Processing Delay:** Real-time processing may cause slight delays.
- **Data Limitations:** Limited dataset may affect model performance.

To manage these risks, proper preprocessing, good quality datasets, and optimized models are used. Testing under different conditions also helps reduce errors.

6.3 PROJECT PLAN

The project plan defines the steps, timeline, and resources required for completing the project.

The development of this project follows these stages:

- Problem identification and requirement analysis
- Data collection (FER2013 dataset)
- Data preprocessing and cleaning
- Model development using CNN
- Model training and testing
- Integration with music recommendation system

- User interface development
- Testing and evaluation
- Final implementation and documentation

Proper planning ensures that the project is completed on time and within scope.

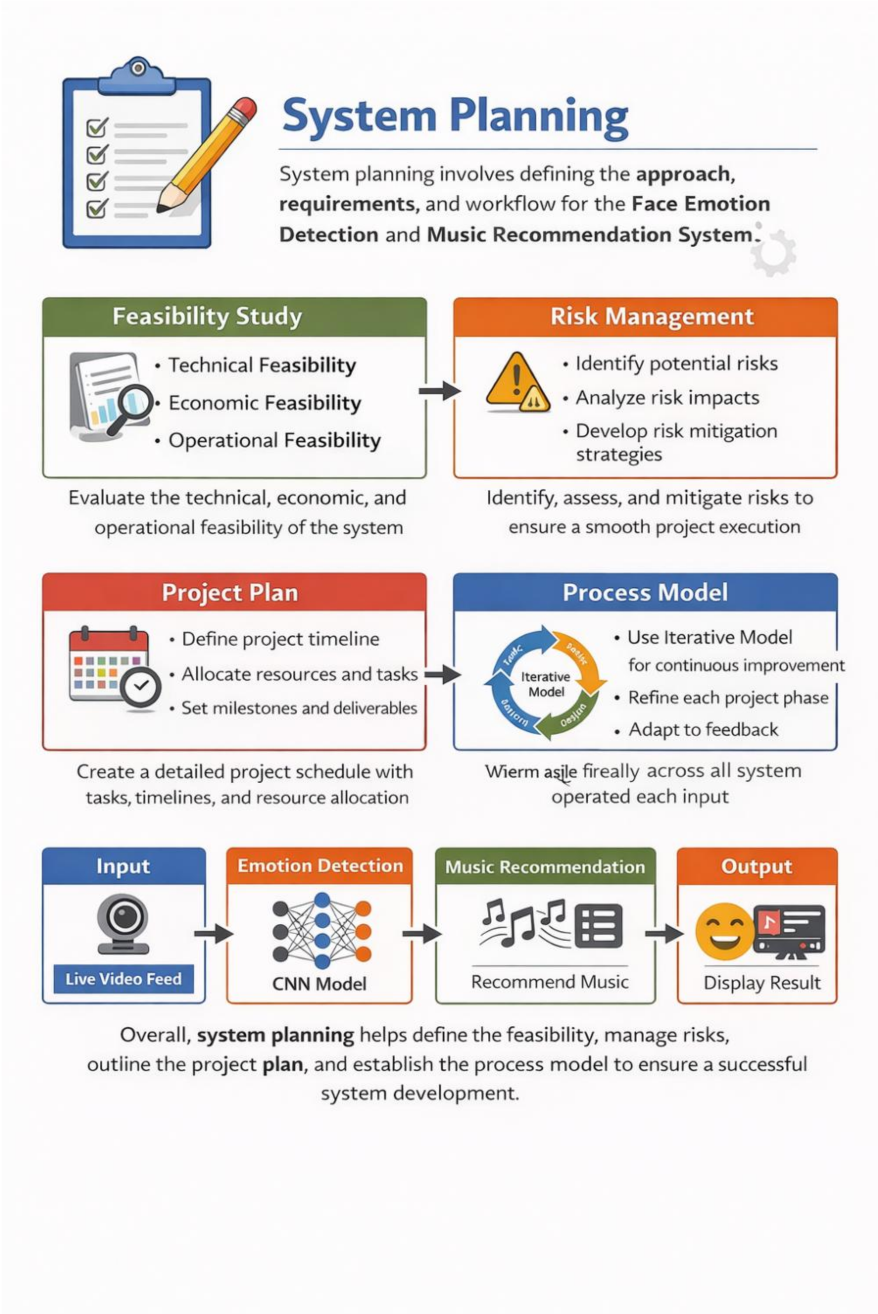
6.4 PROCESS MODEL

The process model defines the development approach followed in the project. This project uses an **Iterative Model**.

In the iterative model, the system is developed in small parts, and each part is tested and improved continuously. The steps include:

- Requirement analysis
- Design
- Implementation
- Testing
- Improvement

This model helps in identifying errors early and improving system performance step by step. It is suitable for this project as it allows continuous refinement of the emotion detection model.


System Planning
System planning involves defining the **approach**, **requirements,** and workflow for the **Face Emotion Detection** and **Music Recommendation System**.
Feasibility Study
Technical Feasibility
Economic Feasibility
Operational Feasibility
Evaluate the technical, economic, and operational feasibility of the system
Risk Management
Identify potential risks
Analyze risk impacts
Develop risk mitigation strategies
Identify, assess, and mitigate risks to ensure a smooth project execution
Project Plan
Define project timeline
Allocate resources and tasks
Set milestones and deliverables
Create a detailed project schedule with tasks, timelines, and resource allocation
Process Model
Iterative Model
Use Iterative Model for continuous improvement
Refine each project phase
Adapt to feedback
Wierm asile fireally across all system operated each input
Input
Live Video Feed
Emotion Detection
CNN Model
Music Recommendation
Recommend Music
Output
Display Result
Overall, **system planning** helps define the feasibility, manage risks, outline the project **plan**, and establish the process model to ensure a successful system development.

SYSTEM REQUIREMENT

The System Requirement Specification (SRS) for the project “Face Emotion Detection and Music Recommendation System” defines the functional and non-functional requirements of the system. It describes what the system will do and the constraints under which it will operate. The system is designed to detect human emotions using facial expressions and recommend suitable music based on the detected emotion.

7.1 SYSTEM REQUIREMENT

Hardware Requirements

The minimum hardware required to run the system efficiently is:

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Hard Disk: At least 10 GB free space
- Webcam: Required for real-time emotion detection
- Keyboard and Mouse
- Display Monitor

Software Requirements

The software environment required for development and execution includes:

- Operating System: Windows / Linux / macOS
- Programming Language: Python
- Libraries/Frameworks:
  - OpenCV (for image processing and face detection)
  - TensorFlow / Keras (for CNN model)
  - NumPy (for numerical computation)
  - Pandas (for data handling)

- Matplotlib (for visualization)

- IDE: Jupyter Notebook / PyCharm / VS Code
- Dataset: FER2013 dataset

7.2 FUNCTIONAL REQUIREMENT

The system must perform the following functions:

- Capture real-time video using webcam
- Detect human face from video frames
- Preprocess facial images (resize, grayscale, normalization)
- Predict emotion using CNN model
- Classify emotions into categories:
  - Happy
  - Sad
  - Angry
  - Neutral
  - Surprise
- Map detected emotion to music playlist
- Recommend suitable songs based on emotion
- Display output on user interface in real time

7.3 NON-FUNCTIONAL REQUIREMENT

- **Performance:** System should process video frames in real time with minimal delay
- **Accuracy:** Emotion detection model should provide high accuracy based on trained dataset
- **Reliability:** System should work consistently under normal lighting conditions
- **Usability:** User-friendly interface with simple interaction
- **Scalability:** System can be upgraded to support more emotions or music platforms

- **Portability:** Can be run on different operating systems with Python support

MODULE SPECIFICATION

The Face Emotion Detection and Music Recommendation System is divided into several interdependent modules. Each module performs a specific function in the pipeline of Machine Learning-based emotion recognition and recommendation.

8.1 Face Extraction & Image Preprocessing

(face_emotion_utils.py)

This module captures each frame of the live webcam feed via OpenCV. It utilizes Haar Cascades or MTCNN to find the bounding boxes around any human face in the camera frame.

Once a face is found, the sub-image is cropped, converted to grayscale, and histogram equalization is applied to normalize lighting. It is then resized to a strictly defined resolution (e.g., 48x48) compatible with the model input.

8.2 Emotion Classification Model

An offline Convolutional Neural Network (CNN) saved in HDF5 format (emotion_model.h5) runs inference on the preprocessed face. The model classifies the face into several emotion labels such as Happy, Sad, Angry, or Neutral, extracting a confidence score for each prediction.

8.3 Audio & Music Recommendation Engine

(spotify_recommender.py & Internal Audio Generator)

Depending on the inferred emotion, the system automatically pulls a relevant song from the local file directory (e.g. static/songs/happy/) or interacts with Spotipy API to recommend top actual tracks. The song URL is dynamically passed back to the frontend without a page reload using an AJAX JSON route.

8.4 Flask Backend server & Web Frontend

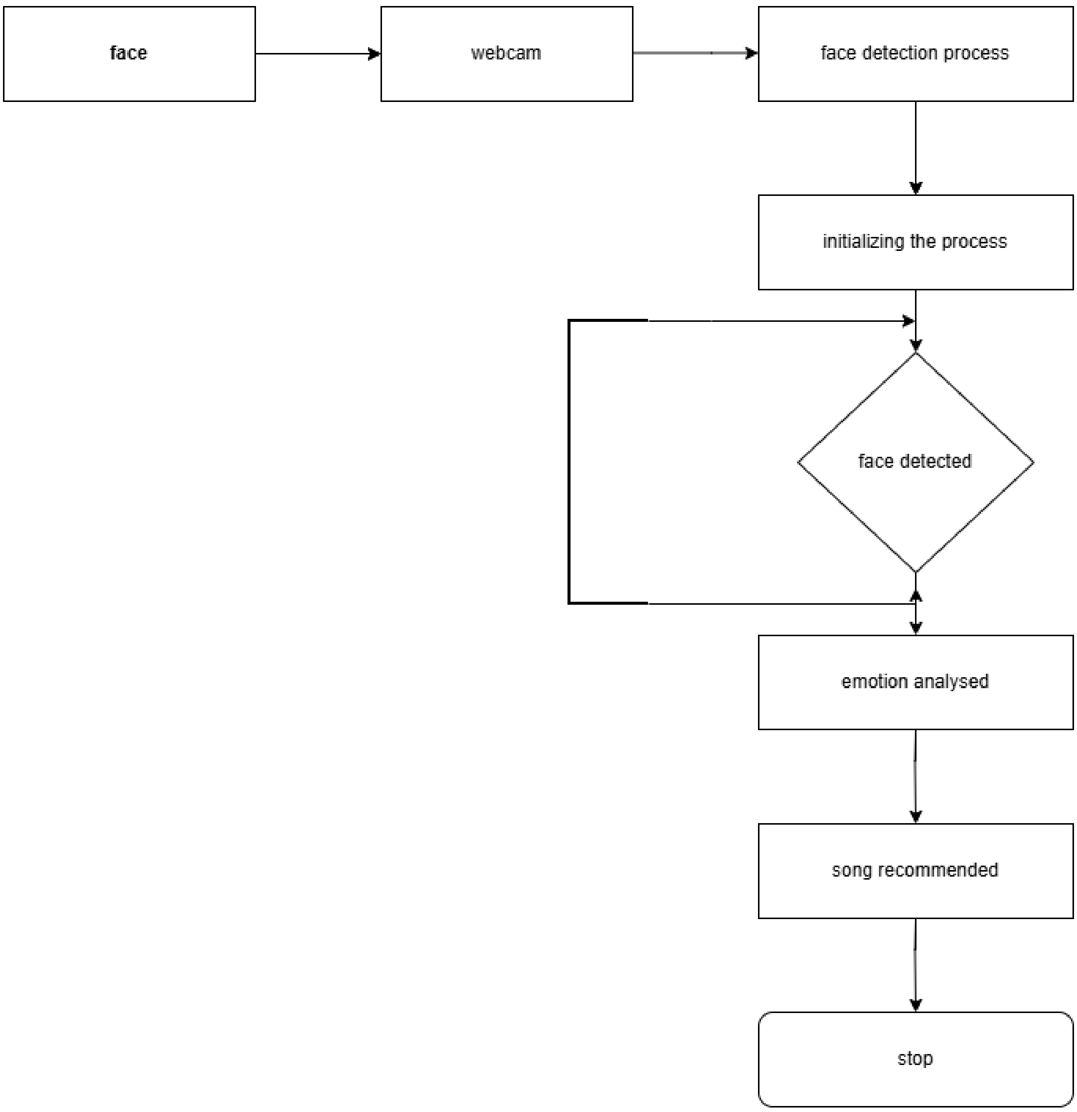
(app.py & index.html)

The Flask server acts as the central command node. It hosts the /video_feed multi-part route for streaming video as images and the /get_emotion route to pass JSON variables synchronously.

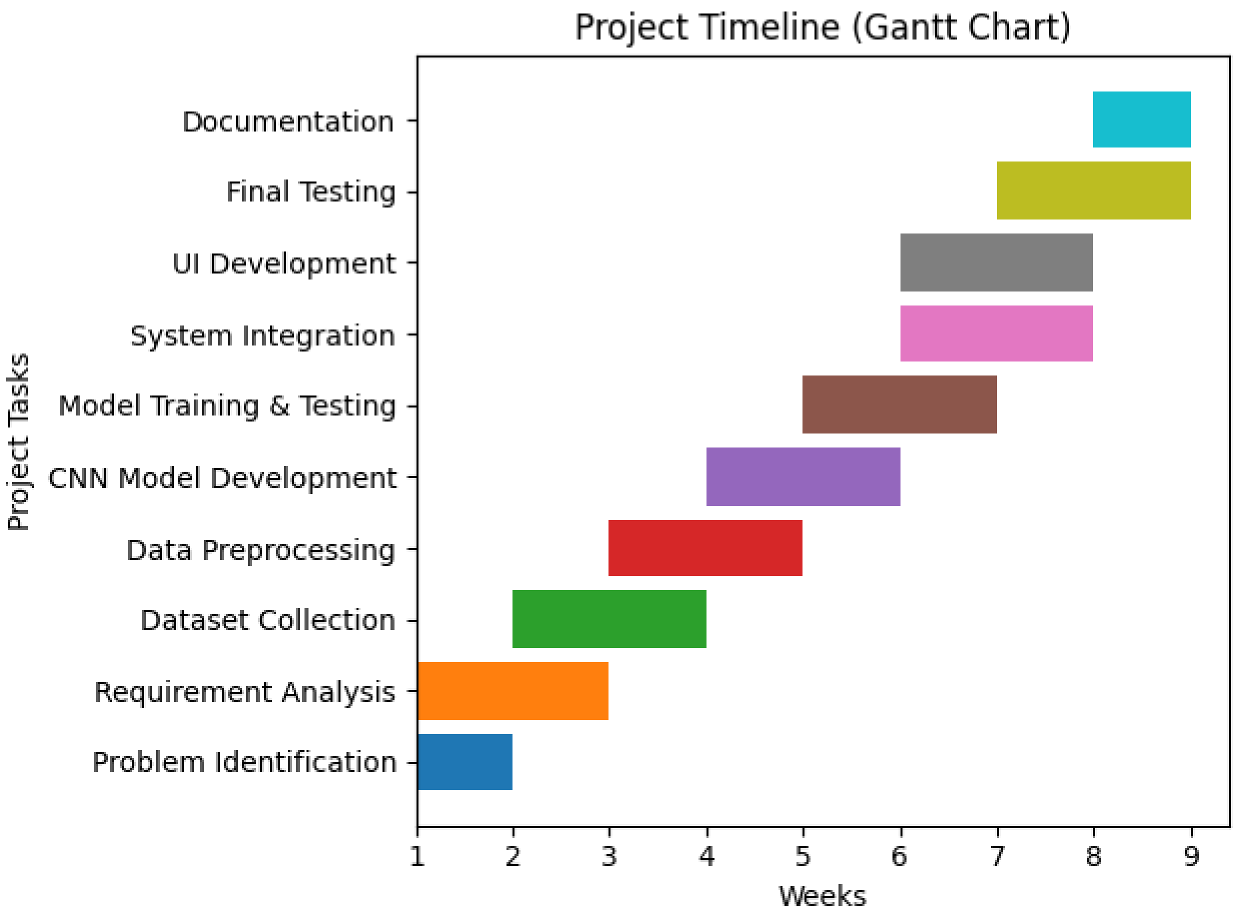
The Web Frontend consumes the JSON variable, updating the text to display the current emotion and confidence alongside directly playing the audio file via the HTML5 <audio> element APIs.

SYSTEM DESIGN

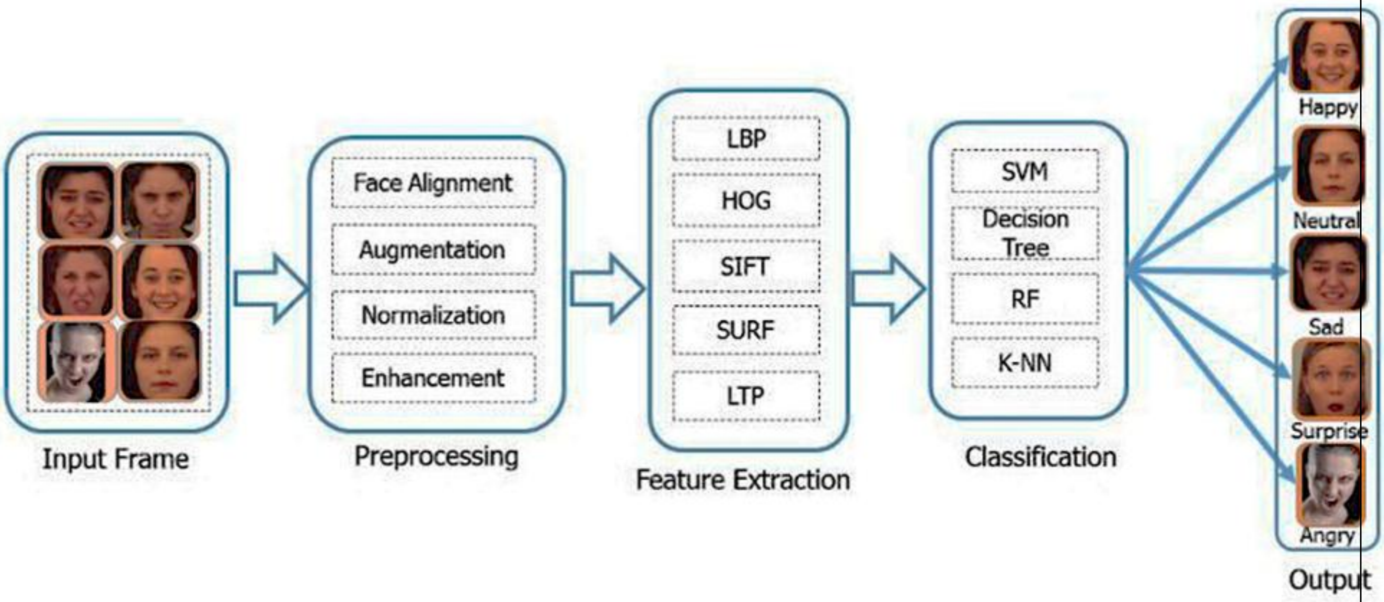
9.1 TASK DEPENDENCY DIAGRAM



9.2 TIME-LINE CHART



9.3 PROJECT FLOW DIAGRAM



STANDARDIZATION OF CODING

Standardization of coding refers to the use of proper coding practices, conventions, and techniques to ensure that the software is efficient, readable, maintainable, and error-free. In the "Face Emotion Detection and Music Recommendation System," standard coding practices are followed to improve performance and reliability.

10.1 Code Efficiency

Code efficiency is important to ensure that the system performs quickly and uses minimal system resources. In this project, the following techniques are used to improve efficiency:

- Efficient use of libraries such as OpenCV and TensorFlow for optimized performance
- Use of pre-trained CNN models to reduce training time
- Avoidance of unnecessary computations and loops
- Image resizing (48×48) to reduce processing time
- Real-time frame processing optimized for speed

Efficient coding ensures that the system can process webcam input in real time without lag.

10.2 Error Handling

Error handling is implemented to ensure that the system works smoothly even when unexpected situations occur.

- Handling of webcam access errors (e.g., camera not available)
- Validation of input images before processing
- Exception handling using try-except blocks in Python
- Handling cases where face is not detected in frame
- Prevention of system crashes due to invalid inputs

Proper error handling improves system stability and user experience.

10.3 Parameter Passing / Calling

Proper parameter passing ensures modular and reusable code.

- Functions are designed with clear input and output parameters
- Example: Passing image frames to preprocessing and CNN model
- Use of function-based modular programming
- Use of arguments for flexibility and scalability

This helps in maintaining clean and structured code.

10.4 Validation Checks

Validation ensures that the input data is correct and suitable for processing.

- Checking if image is properly captured from webcam
- Ensuring correct image size before feeding into CNN model
- Validating pixel normalization values
- Checking dataset integrity during training
- Ensuring valid emotion output before recommendation

Validation checks help in improving accuracy and preventing errors.

10.5 Code Readability and Maintainability

Readable and maintainable code is essential for future improvements.

- Use of meaningful variable and function names
- Proper indentation and formatting
- Use of comments to explain important code segments
- Modular structure (separate functions for each module)
- Following Python coding standards (PEP guidelines)

This makes the code easy to understand and modify.

10.6 Reusability

- Functions are reusable across different modules
- Modular design allows easy integration of new features

- Code can be extended for mobile or web applications

10.7 Documentation of Code

- Inline comments are added in the code
- Each function is properly described
- Important logic is explained for better understanding

TESTING

Testing is a crucial phase in the development of the "Face Emotion Detection and Music Recommendation System." It ensures that the system functions correctly, meets the requirements, and performs efficiently under different conditions. Proper testing helps in identifying errors, improving system accuracy, and enhancing overall performance.

11.1 TESTING TECHNIQUES

1. Unit Testing

Unit testing is performed to test individual modules of the system separately. Each component is verified independently to ensure that it works correctly before integrating it with other modules.

In this project, the following modules are tested individually:

- **Input Module:** Testing webcam functionality and frame capture
- **Face Detection Module:** Checking whether faces are correctly detected using OpenCV
- **Preprocessing Module:** Verifying grayscale conversion, resizing, and normalization
- **CNN Model Module:** Testing emotion prediction accuracy
- **Music Recommendation Module:** Ensuring correct mapping of emotion to songs

Unit testing helps in identifying bugs at an early stage and ensures that each module performs its intended function accurately.

2. System Testing

System testing is performed after integrating all modules to verify the complete system functionality. It ensures that the entire workflow—from capturing the image to recommending music—works smoothly.

In this project, system testing includes:

- Testing real-time emotion detection using webcam

- Verifying correct classification of emotions (Happy, Sad, Angry, Neutral, Surprise)
- Checking the accuracy of music recommendations
- Ensuring smooth interaction between modules
- Testing system performance under different lighting conditions

System testing ensures that the system meets user requirements and performs reliably in real-world scenarios.

11.2 TESTING PLAN

A structured testing plan is followed to ensure systematic evaluation of the system.

Step 1: Test Modules Individually

Each module is tested independently using sample inputs. Errors are identified and fixed before moving to the next stage.

Step 2: Integrate Modules

After successful unit testing, all modules are integrated step by step. Integration testing ensures proper communication between modules.

Step 3: Perform Final Testing

The complete system is tested using real-time inputs. This includes checking:

- End-to-end system performance
- Response time
- Accuracy of emotion detection
- Correctness of music recommendation

The testing plan ensures that the system is stable, accurate, and ready for deployment.

11.3 TEST REPORTS

Test reports are generated to evaluate system performance and accuracy.

1. Accuracy Results

- Accuracy of CNN model is calculated using test dataset

- Accuracy indicates how correctly emotions are classified
- Higher accuracy means better performance

2. Confusion Matrix

- A confusion matrix is used to evaluate classification performance
- It shows correct and incorrect predictions for each emotion class
- Helps in identifying which emotions are misclassified

Example:

- Happy → correctly predicted
- Sad → sometimes misclassified as neutral

3. Error Logs

- Records of system errors during execution
- Helps in debugging issues
- Includes errors like:
  - Face not detected
  - Invalid input
  - Model prediction failure

Test reports provide a clear understanding of system strengths and weaknesses.

11.4 DEBUGGING AND OPTIMIZATION

Debugging is the process of identifying and fixing errors in the system. Optimization improves system performance and efficiency.

1. Fixing Model Errors

- Improve dataset quality
- Retrain CNN model
- Adjust hyperparameters

2. Improving Performance

- Optimize image preprocessing

- Reduce model complexity
- Use efficient algorithms

3. Speed Optimization

- Reduce frame processing time
- Use smaller image size (48×48)
- Optimize real-time detection

Debugging and optimization ensure that the system performs efficiently with high accuracy and minimal delay.

SYSTEM SECURITY MEASURES

System security is an important aspect of the "Face Emotion Detection and Music Recommendation System." It ensures that the data used in the system is protected, unauthorized access is prevented, and the system operates in a secure and reliable manner. Since the system processes user facial data and real-time inputs, proper security measures are essential to maintain privacy and integrity.

12.1 DATA SECURITY

Data security refers to the protection of data from unauthorized access, misuse, or loss. In this project, the following measures are implemented to ensure data security:

1. Local Storage of Data

- All data used in the system, including the dataset (FER2013) and music mapping files, is stored locally on the system.
- No cloud storage or external servers are used, reducing the risk of data breaches.
- Local storage ensures faster data access and better control over data handling.

2. No External Data Sharing

- The system does not share user data with any third-party applications or services.
- Facial images captured through the webcam are processed in real time and are not stored permanently.
- This ensures user privacy and prevents misuse of sensitive data.

3. Secure Handling of Input Data

- Input images are processed temporarily and deleted after processing.
- No unnecessary data storage is performed.
- This minimizes the chances of data leakage.

4. Data Integrity

- The dataset used for training is verified and properly labeled.
- Data corruption is avoided through validation checks.

- Ensures accurate and reliable system performance.

5. Backup and Recovery (Optional)

- Important files such as trained models and datasets can be backed up securely.
- In case of system failure, data can be restored without loss.

12.2 USER ACCESS CONTROL

User access control ensures that only authorized users can access and operate the system. It also prevents misuse and unauthorized manipulation of system data.

1. Basic Access Control

- The system is designed for controlled usage on a local machine.
- Only authorized users (e.g., developer or operator) can run the application.
- Access to system files and code is restricted.

2. Secure System Usage

- The system does not require sensitive user information such as passwords or personal details.
- Minimal user input reduces security risks.
- The application runs in a controlled environment, ensuring safe operation.

3. File Access Protection

- Important project files such as model weights and datasets are protected from unauthorized editing.
- File permissions can be set to restrict access.

4. Prevention of Unauthorized Access

- The system can be protected using OS-level security (login credentials).
- Unauthorized users cannot access the system without permission.

5. Safe Execution Environment

- The application is executed in a secure environment (local machine or trusted system).
- Avoids risks associated with online deployment.

12.3 ADDITIONAL SECURITY ENHANCEMENTS

To further strengthen the system, the following advanced security features can be implemented:

- User authentication system (username and password)
- Data encryption for stored files
- Secure API integration (for music streaming platforms)
- Role-based access control (admin/user roles)
- Secure logging and monitoring mechanisms

REPORTS

Reports are an essential part of the "Face Emotion Detection and Music Recommendation System." They provide a clear representation of the system's output, performance, and functionality. Reports help users and developers understand how the system behaves in real-time and evaluate its effectiveness.

The system generates different types of reports based on emotion detection and music recommendation results.

13.1 DETECTED EMOTION REPORT

The detected emotion report displays the emotional state of the user identified by the system through facial expression analysis.

Features of Emotion Report:

- Displays real-time emotion detected from webcam input
- Shows emotion label such as:
  - Happy
  - Sad
  - Angry
  - Neutral
  - Surprise
- Emotion is updated continuously as facial expressions change
- Provides visual feedback on the screen along with webcam feed

Example Output:

- User face detected → Emotion: **Happy**
- User face detected → Emotion: **Sad**

Importance:

- Helps in understanding user mood in real time
- Acts as input for music recommendation module

- Improves interaction between user and system

13.2 RECOMMENDED MUSIC REPORT

This report provides a list of songs or music categories based on the detected emotion.

Features of Music Report:

- Displays recommended songs based on detected emotion
- Maps emotions to music categories:
  - Happy → Energetic / Party songs
  - Sad → Emotional / Soft songs
  - Angry → Calm / Relaxing music
  - Neutral → Mixed playlist
  - Surprise → Trending songs
- Shows multiple song suggestions
- Can include song names, artists, or playlist links

Example Output:

- Emotion: Happy → Songs: Party songs, Dance music
- Emotion: Sad → Songs: Slow songs, Emotional music

Importance:

- Provides personalized user experience
- Saves time in searching songs manually
- Enhances user satisfaction

13.3 COMBINED OUTPUT REPORT

The system also provides a combined output where both emotion and music recommendation are displayed together.

Features:

- Real-time webcam display
- Emotion label on screen
- Suggested songs list
- Continuous updates

Example:

Emotion	**Recommended Music**
Happy	Party Songs
Sad	Soft Music

13.4 REPORT FORMAT

Reports can be presented in different formats:

- On-screen display (GUI output)
- Text-based logs
- Tables and charts
- Screenshots for documentation

APPENDICES

Appendices are supplementary sections of the project documentation that include additional information supporting the main report. These materials provide detailed insights into the implementation, execution, and results of the system. They help in better understanding of the project and serve as reference material.

14.1 SOURCE CODE

The source code is an essential part of the appendix, as it represents the actual implementation of the system.

Description:

- The project is implemented using Python programming language
- Major libraries used include:
  - OpenCV (for face detection)
  - TensorFlow / Keras (for CNN model)
  - NumPy and Pandas (for data handling)
- The code is modular, with separate files/functions for:
  - Face detection
  - Image preprocessing
  - Emotion classification
  - Music recommendation

Features of Source Code:

- Well-structured and organized
- Proper use of functions and modules
- Includes comments for better understanding
- Follows standard coding practices

Example Snippet:

```
import cv2
from tensorflow.keras.models import load_model

model = load_model('emotion_model.hdf5')

def detect_emotion(frame):
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    face = cv2.resize(gray, (48,48))
    face = face / 255.0
    face = face.reshape(1,48,48,1)
    prediction = model.predict(face)
    return prediction
```

Importance:

- Helps evaluators understand implementation
- Useful for debugging and future enhancements
- Demonstrates practical knowledge

14.2 OUTPUT SCREENSHORTS

Output screenshots provide visual proof of system functionality and results.

Description:

- Screenshots are captured during system execution
- They show real-time working of the system

Includes:

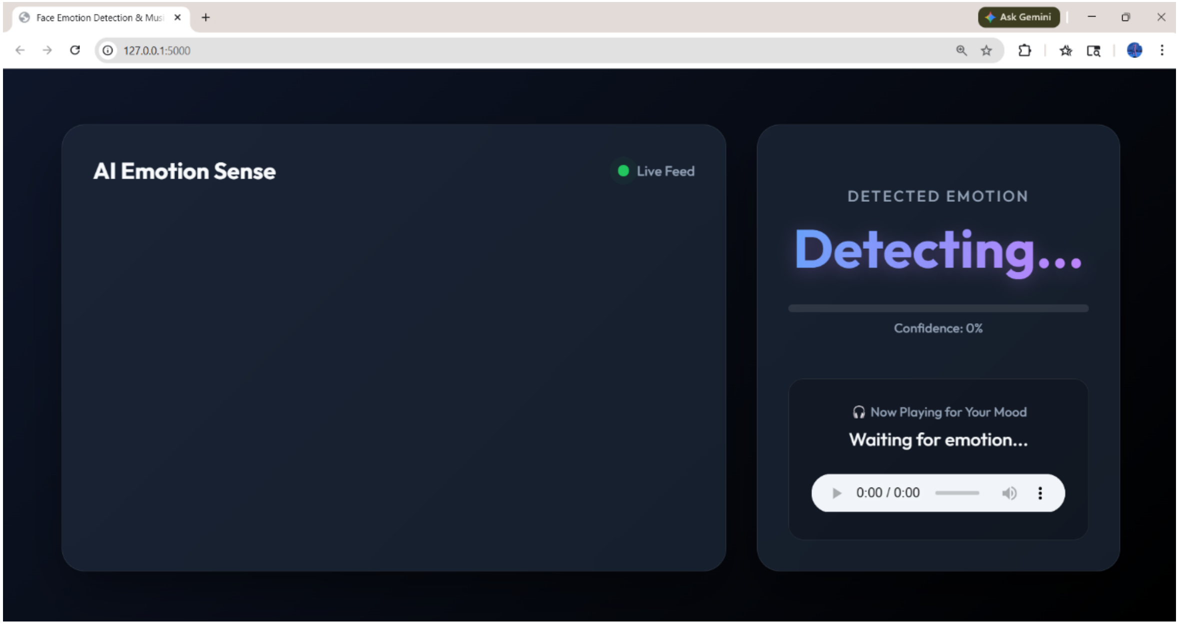
- Webcam feed with detected face
- Emotion label displayed on screen
- Recommended music list
- GUI interface of the system

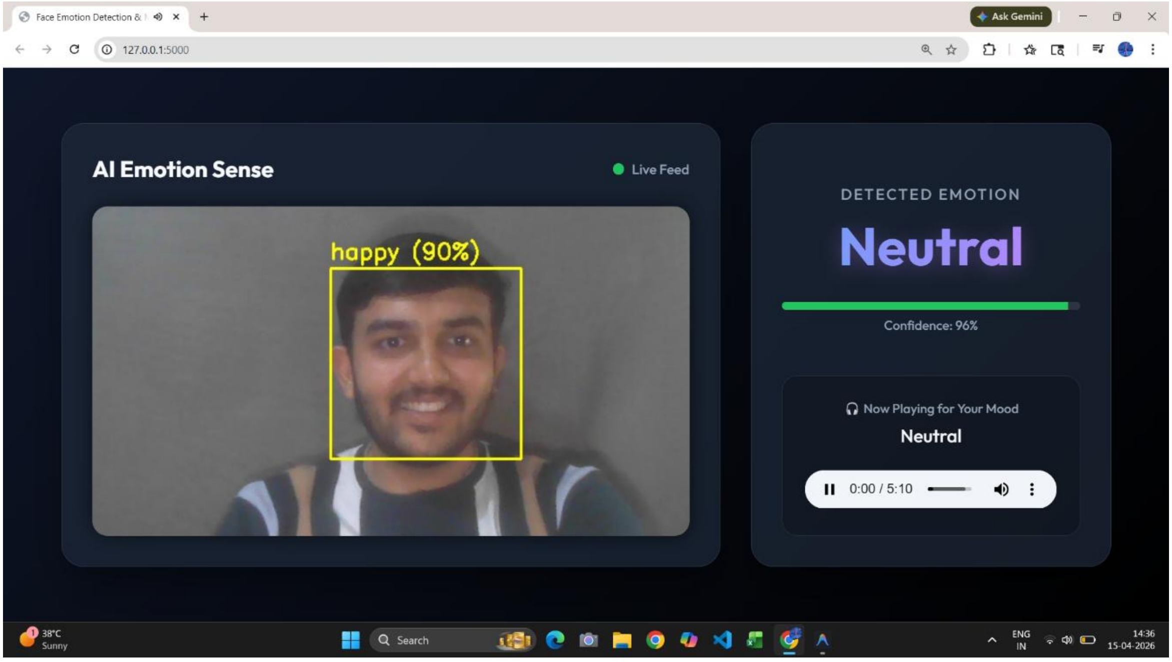
Example Cases:

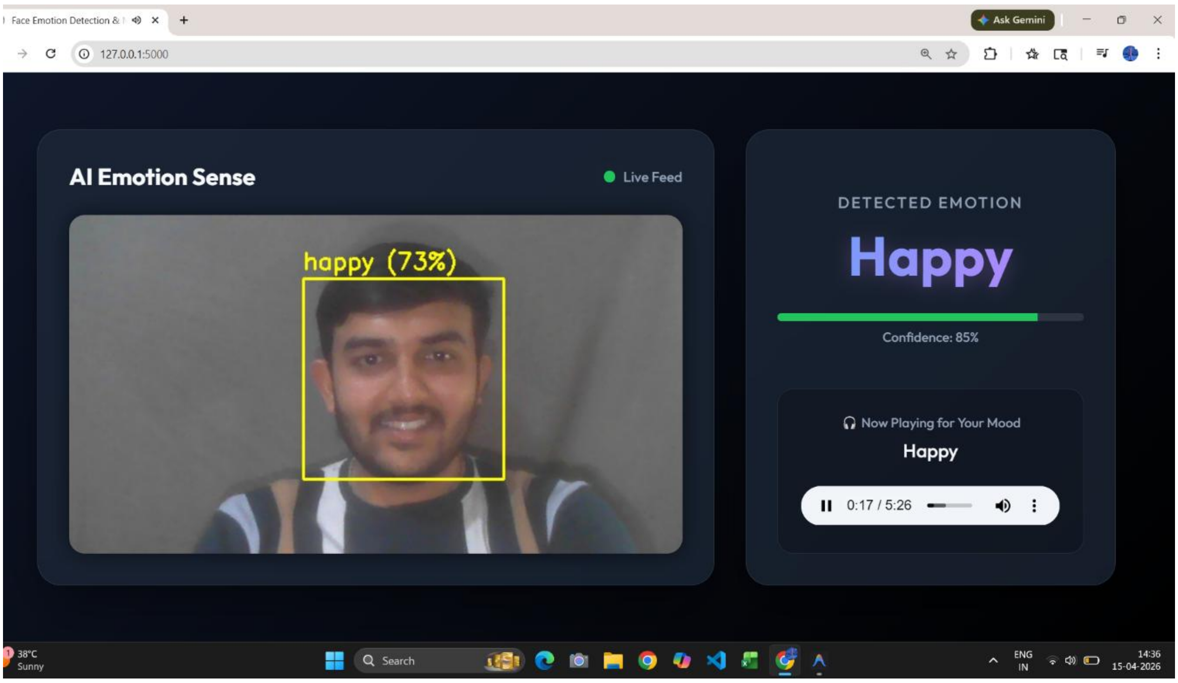
- Screenshot showing "Happy" emotion with party songs
- Screenshot showing "Sad" emotion with soft music

Importance:

- Provides visual validation of system
- Helps in explaining project during viva
- Improves presentation of report

- 

- 
Face Emotion Detection &
Ask Gemini
127.0.0.1:5000
AI Emotion Sense
Live Feed
happy (90%)
DETECTED EMOTION
Neutral
Confidence: 96%
Now Playing for Your Mood
Neutral
0:00 / 5:10
38°C
Sunny
Search
ENG
IN
14:36
15-04-2026

- 
Face Emotion Detection &
Ask Gemini
127.0.0.1:5000
AI Emotion Sense
Live Feed
happy (73%)
DETECTED EMOTION
Happy
Confidence: 85%
Now Playing for Your Mood
Happy
0:17 / 5:26
38°C
Sunny
Search
ENG
IN
14:36
15-04-2026

14.3 MODEL TRAINING GRAPHS

Model training graphs are included to show the performance of the CNN model during training and testing.

Description:

- Graphs are generated during model training phase
- Represent learning progress of the model

Types of Graphs:

1. **Accuracy Graph**
    - Shows training accuracy vs validation accuracy
    - Indicates how well the model is learning
2. **Loss Graph**
    - Shows training loss vs validation loss
    - Helps identify overfitting or underfitting

Example Interpretation:

- Increasing accuracy → Model is improving
- Decreasing loss → Model errors are reducing
- Large gap between training and validation → Overfitting

Importance:

- Helps evaluate model performance
- Useful for improving model accuracy
- Demonstrates understanding of Machine Learning concepts

GLOSSARY

The glossary provides definitions of important technical terms used in the project.

1. CNN (Convolutional Neural Network)

CNN is a deep learning algorithm used for image processing and classification. It automatically extracts features from images and is used in this project to detect emotions from facial expressions.

2. AI (Artificial Intelligence)

AI refers to machines that can perform tasks requiring human intelligence such as learning, decision-making, and problem-solving. This project uses AI to understand user emotions and recommend music.

3. ML (Machine Learning)

Machine Learning is a part of AI that allows systems to learn from data and improve performance without being explicitly programmed. It is used here to train the emotion detection model.

4. OpenCV (Open Source Computer Vision Library)

OpenCV is a library used for image and video processing. In this project, it is used for capturing webcam input and detecting faces.

5. Deep Learning

Deep Learning is a type of Machine Learning that uses neural networks with multiple layers. CNN used in this project is a deep learning model.

6. Computer Vision

Computer Vision enables machines to understand visual data like images and videos. It is used for face detection and emotion analysis.

7. Dataset

A dataset is a collection of data used for training models. This project uses the FER2013 dataset for emotion classification.

8. Emotion Detection

Emotion detection is the process of identifying human emotions from facial expressions. This is the main function of the system.

9. Real-Time Processing

Real-time processing means analyzing data instantly as it is received. The system detects emotions and recommends music instantly.

10. Preprocessing

Preprocessing is the process of preparing data before using it in the model, such as resizing and normalizing images.